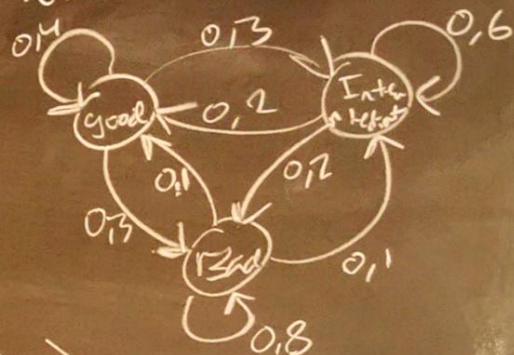


HMM-Shit - discrete

To model uncertainty of System State (Faulty)
Use HMM. Since we can only use observations
Rx is Packet Rx correctly, what is wireless link state
 $\uparrow$
 $e = \text{error}$
 $c = \text{correct}$



HMM fully described by:

$$\text{HMM} := \langle E, V, \Pi, \underline{P}, \underline{B} \rangle$$

E : Hidden State Space (good, Intermediate, poor)

V : Set of possible observation symbols (Correct, error)

Π : Init State Probability @ Step 1

$\underline{P}$: $N \times N$ Matrix State transition Probabilities

$$P_{ij} = \Pr(X_{k+1}=j | X_k=i) \quad \text{Hidden State}$$

$\underline{B}$: $N \times M$ Matrix Symbol generation Probabilities
(Prob. obs state c given hidden state good)

$$b_{ij} = \Pr(O_k=j | X_k=i)$$